

Exploratory Factor Analysis

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```
rm(list=ls())  
# install.packages("GPArotation")  
library("readxl")  
library("psych")  
library(dplyr)
```

```
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':
```

```
##
##      filter, lag

## The following objects are masked from 'package:base':
##
##      intersect, setdiff, setequal, union

library(tidyr)
library(bruceR)

##
## bruceR (v0.8.10)
## BRoadly Useful Convenient and Efficient R functions
##
## Packages also loaded:
##   data.table   emmeans
##   dplyr        lmerTest
##   tidyr        effectsize
##   stringr      performance
##   ggplot2      interactions
##
## Main functions of `bruceR`:
## cc()           Describe()  TTEST()
## add()          Freq()      MANOVA()
## .mean()        Corr()      EMMEANS()
## set.wd()       Alpha()     PROCESS()
## import()       EFA()       model_summary()
## print_table()  CFA()       lavaan_summary()
##
## For full functionality, please install all dependencies:
## install.packages("bruceR", dep=TRUE)
##
## Online documentation:
## https://psychbruce.github.io/bruceR
```

```
##
## NEWS: A new version of bruceR (2026.1) is available (2026-01-29)!
##
## ***** Update *****
## install.packages("bruceR", dep=TRUE)
## *****

library(corrplot)

## corrplot 0.92 loaded

library(ggplot2)
library(GPARotation)

##
## Attaching package: 'GPARotation'

## The following objects are masked from 'package:psych':
##
##      equamax, varimin

currentData = psych::bfi
currentData = currentData %>% drop_na()
currentData = currentData[complete.cases(currentData),]
currentData = currentData[1:25]

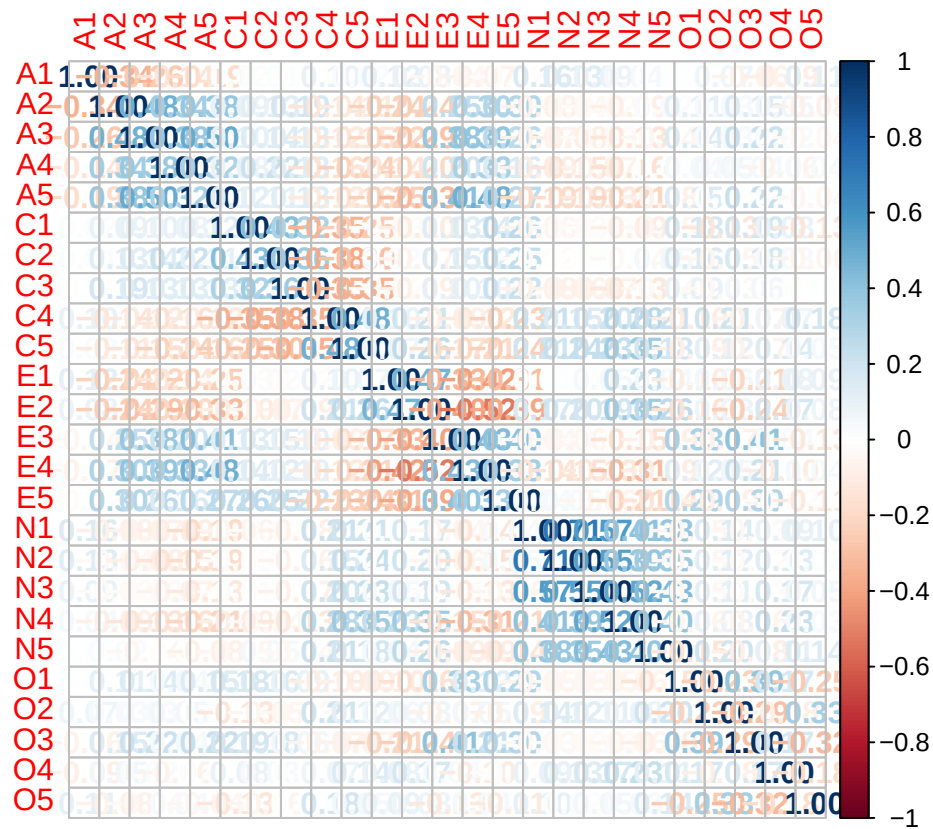
describe(currentData)
```

##	vars	n	mean	sd	median	trimmed	mad	min	max	range	skew	kurtosis	se
##	A1	1	2236	2.37	1.39	2	2.17	1.48	1	6	5	0.88	-0.17 0.03
##	A2	2	2236	4.83	1.16	5	5.01	1.48	1	6	5	-1.15	1.14 0.02
##	A3	3	2236	4.63	1.29	5	4.82	1.48	1	6	5	-1.03	0.56 0.03
##	A4	4	2236	4.75	1.45	5	4.99	1.48	1	6	5	-1.09	0.23 0.03
##	A5	5	2236	4.58	1.26	5	4.74	1.48	1	6	5	-0.88	0.24 0.03
##	C1	6	2236	4.57	1.22	5	4.71	1.48	1	6	5	-0.89	0.42 0.03
##	C2	7	2236	4.40	1.31	5	4.54	1.48	1	6	5	-0.77	-0.09 0.03
##	C3	8	2236	4.32	1.29	5	4.44	1.48	1	6	5	-0.69	-0.11 0.03

## C4	9	2236	2.50	1.36	2	2.35	1.48	1	6	5	0.64	-0.56	0.03
## C5	10	2236	3.26	1.63	3	3.20	1.48	1	6	5	0.09	-1.23	0.03
## E1	11	2236	2.97	1.62	3	2.86	1.48	1	6	5	0.38	-1.07	0.03
## E2	12	2236	3.12	1.61	3	3.04	1.48	1	6	5	0.25	-1.13	0.03
## E3	13	2236	4.01	1.34	4	4.08	1.48	1	6	5	-0.48	-0.43	0.03
## E4	14	2236	4.43	1.46	5	4.60	1.48	1	6	5	-0.85	-0.27	0.03
## E5	15	2236	4.42	1.33	5	4.57	1.48	1	6	5	-0.81	-0.03	0.03
## N1	16	2236	2.91	1.56	3	2.80	1.48	1	6	5	0.39	-0.99	0.03
## N2	17	2236	3.49	1.53	4	3.48	1.48	1	6	5	-0.06	-1.07	0.03
## N3	18	2236	3.20	1.60	3	3.14	1.48	1	6	5	0.17	-1.18	0.03
## N4	19	2236	3.18	1.56	3	3.11	1.48	1	6	5	0.22	-1.06	0.03
## N5	20	2236	2.95	1.62	3	2.83	1.48	1	6	5	0.40	-1.05	0.03
## O1	21	2236	4.82	1.12	5	4.97	1.48	1	6	5	-0.91	0.47	0.02
## O2	22	2236	2.69	1.55	2	2.54	1.48	1	6	5	0.61	-0.77	0.03
## O3	23	2236	4.48	1.19	5	4.60	1.48	1	6	5	-0.79	0.40	0.03
## O4	24	2236	4.95	1.18	5	5.15	1.48	1	6	5	-1.26	1.26	0.02
## O5	25	2236	2.46	1.33	2	2.30	1.48	1	6	5	0.78	-0.16	0.03

0.1 Correlation Matrix

```
cor_matrix = cor(currentData)
corrplot(cor_matrix, method="number")
```



0.2 Data Adequacy

get the overall MSA from KMO; look at the bartlett test

```
KMO(cor(currentData))
```

```
## Kaiser-Meyer-Olkin factor adequacy
```

```
## Call: KMO(r = cor(currentData))
```

```
## Overall MSA = 0.85
```

```
## MSA for each item =
```

```
##   A1   A2   A3   A4   A5   C1   C2   C3   C4   C5   E1   E2   E3   E4   E5   N1
## 0.74 0.83 0.87 0.87 0.90 0.84 0.79 0.85 0.82 0.86 0.84 0.88 0.89 0.88 0.89 0.78
##   N2   N3   N4   N5   O1   O2   O3   O4   O5
## 0.78 0.86 0.89 0.86 0.86 0.78 0.83 0.78 0.76
```

```
cortest.bartlett(cor(currentData),nrow(currentData))
```

```
## $chisq  
## [1] 16484.78  
##  
## $p.value  
## [1] 0  
##  
## $df  
## [1] 300
```

```
det(cor_matrix)
```

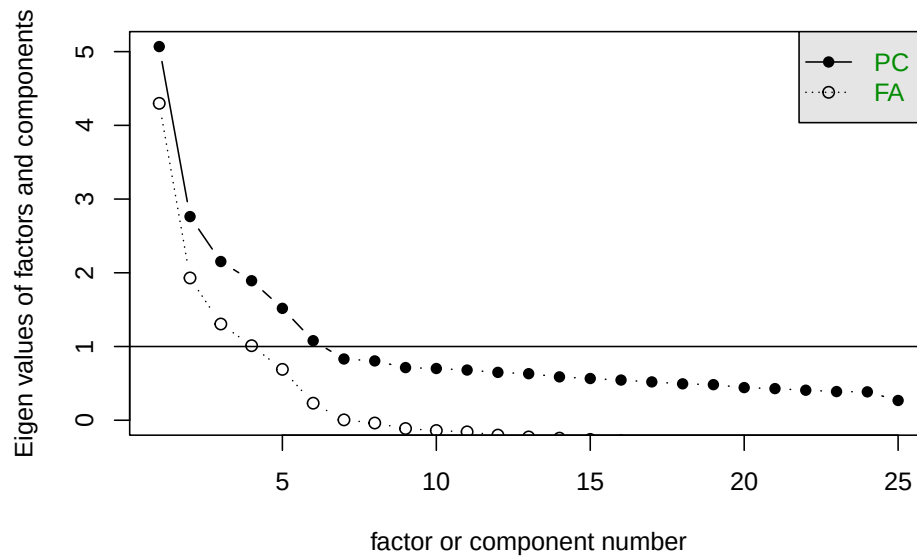
```
## [1] 0.0006075253
```

0.3 Determine num. of Factors

- To determine the number of factors, scree and parallel are two methods.
- Scree is obsolete; parallel suggests 3 factors here.

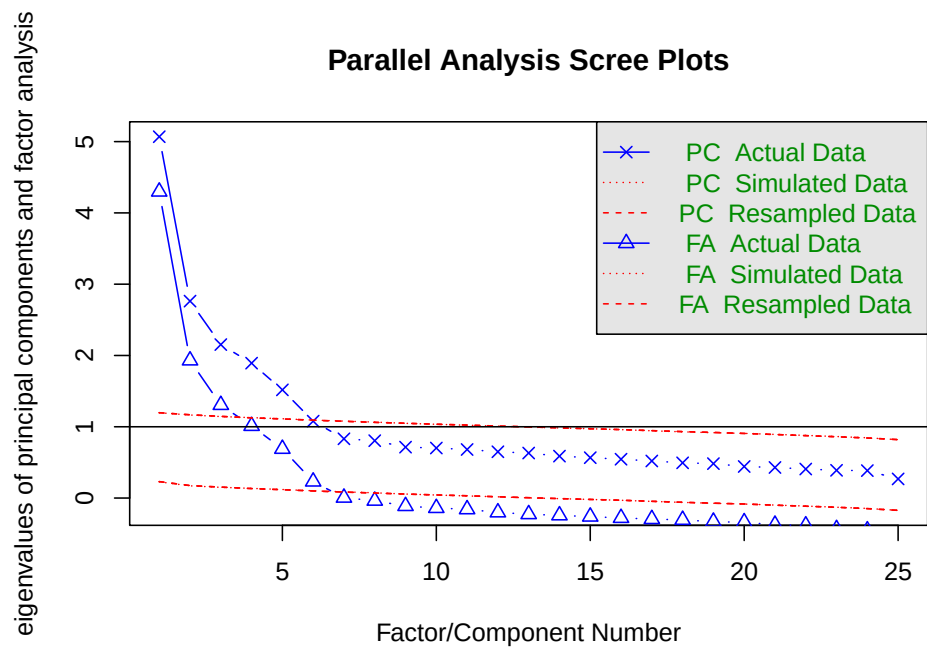
```
scree(currentData)
```

Scree plot



```
fa.parallel(currentData, fm='pa', fa='both')
```

Parallel Analysis Scree Plots



Parallel analysis suggests that the number of factors = 6 and the number of components = 6

1 Factor Analysis

1.1 NULL Model

Take a look: without selecting the number of factors, without rotation.

Factoring Method “**fm**”: * **minres**: do a minimum residual * **pa**: do the principal factor solution * **ml**: do a maximum likelihood factor analysis

The Proportion Var and Cumulative Var are the variance that can be explained by factors.

“SS loadings” is the sum of squared loadings for each factor; **>1** means the factor is worth keeping.

The model can be printed with the following results:

- Standardized loadings (pattern matrix) based upon correlation matrix
 - SS loadings
 - * The eigenvalues, the sum of the squared loadings for each factor.
 - * **Better to be >1.**
 - Proportion Var
 - * The overall variance (for all variables/items) that can be explained by each factor.
 - Cumulative Var
 - Proportion Explained
 - * The relative importance of each factor.
 - Cumulative Proportion
- Test of the hypothesis that selected numbers of factors are sufficient
- Measures of factor score adequacy
 - Correlation of (regression) scores with factors
 - Multiple R square of scores with factors
 - Minimum correlation of possible factor scores
- Data frame with loading matrix, communality, uniqueness, and item complexity.

Standardized loadings (pattern matrix) based upon correlation

matrix is the most important part!

```
fa_none = fa(currentData,nfactors = ncol(currentData),
             fm='minres',rotate='none')
print(fa_none)
```

```
## Factor Analysis using method = minres
## Call: fa(r = currentData, nfactors = ncol(currentData), rotate = "none",
##      fm = "minres")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      MR1  MR2  MR3  MR4  MR5  MR6  MR7  MR8  MR9  MR10  MR11  MR12
## A1 -0.22 -0.03  0.15  0.03 -0.42  0.31  0.07 -0.05  0.02  0.01 -0.02  0.10
## A2  0.47  0.31 -0.20  0.14  0.37 -0.23  0.14  0.15 -0.12  0.01 -0.05  0.07
## A3  0.53  0.31 -0.25  0.10  0.28  0.02  0.13 -0.12  0.09  0.00 -0.02 -0.12
## A4  0.41  0.13 -0.16  0.30  0.18  0.03 -0.01 -0.23 -0.06  0.23 -0.02  0.17
## A5  0.57  0.19 -0.27  0.04  0.18  0.15  0.02 -0.05  0.19 -0.10  0.07 -0.07
## C1  0.33  0.13  0.47  0.14  0.00  0.11 -0.09  0.19  0.17  0.03 -0.03  0.03
## C2  0.33  0.19  0.49  0.32  0.06  0.19 -0.10  0.06 -0.05  0.23 -0.03 -0.11
## C3  0.33  0.05  0.35  0.34  0.03  0.02  0.07  0.14 -0.07 -0.20  0.06  0.01
## C4 -0.47  0.13 -0.45 -0.24  0.02  0.27  0.08  0.06 -0.11  0.04  0.05  0.04
## C5 -0.50  0.16 -0.29 -0.31  0.12  0.08 -0.01  0.26  0.10  0.17 -0.08 -0.06
## E1 -0.43 -0.21  0.27  0.14  0.27  0.25  0.23 -0.14  0.06 -0.01 -0.09 -0.03
## E2 -0.63 -0.06  0.22  0.09  0.33  0.11  0.11 -0.01  0.03  0.03  0.20  0.02
## E3  0.55  0.34 -0.09 -0.20 -0.10  0.21  0.03 -0.09 -0.07 -0.11  0.01 -0.07
## E4  0.62  0.19 -0.34  0.08 -0.19  0.15 -0.16  0.03  0.15  0.01  0.02  0.12
## E5  0.53  0.30  0.11 -0.04 -0.23 -0.03  0.22  0.15 -0.11  0.02 -0.06  0.02
## N1 -0.43  0.65  0.06  0.09 -0.27 -0.13  0.13 -0.12  0.01  0.05  0.05 -0.03
## N2 -0.42  0.65  0.12  0.05 -0.22 -0.21  0.15  0.00  0.14  0.05  0.09 -0.01
## N3 -0.40  0.63  0.06  0.06 -0.01 -0.01 -0.13 -0.10  0.03 -0.11 -0.16  0.05
## N4 -0.54  0.42  0.09 -0.04  0.23  0.08 -0.11 -0.02 -0.06 -0.08 -0.22 -0.02
## N5 -0.36  0.44 -0.05  0.22  0.13  0.05 -0.27  0.02 -0.15 -0.03  0.18 -0.01
## O1  0.33  0.18  0.24 -0.36  0.02  0.16  0.14 -0.01 -0.09  0.01 -0.02  0.10
## O2 -0.20  0.09 -0.33  0.39 -0.05  0.15  0.10  0.15 -0.01 -0.01  0.01  0.01
```

```

## 03  0.40  0.28  0.19 -0.47  0.03  0.13 -0.05 -0.03 -0.08  0.05  0.11 -0.11
## 04 -0.09  0.24  0.18 -0.24  0.31  0.07  0.01  0.08  0.08 -0.08  0.06  0.18
## 05 -0.20 -0.05 -0.29  0.44 -0.15  0.21  0.06  0.06 -0.07 -0.05 -0.02 -0.06
##      MR13  MR14  MR15  MR16  MR17  MR18  MR19  MR20  MR21  MR22  MR23  MR24  MR25
## A1  0.05  0.17  0.00  0.08 -0.02 -0.02 -0.03  0.03  0.02 -0.01  0.00   0   0
## A2  0.07  0.07 -0.07  0.07  0.00  0.05 -0.01  0.02  0.00  0.03  0.00   0   0
## A3  0.01 -0.02  0.06  0.02 -0.08  0.01 -0.06  0.02  0.00 -0.05  0.00   0   0
## A4  0.02  0.02  0.07  0.00 -0.01 -0.03  0.04  0.00  0.00  0.00  0.01   0   0
## A5  0.04  0.09 -0.03 -0.05  0.06 -0.09  0.01 -0.03  0.03  0.02  0.00   0   0
## C1  0.08 -0.03 -0.03 -0.07 -0.13  0.00 -0.01  0.01 -0.04  0.01  0.01   0   0
## C2 -0.08 -0.02  0.01  0.03  0.04  0.00  0.02 -0.03  0.01  0.00 -0.01   0   0
## C3  0.09  0.01  0.12  0.12  0.02 -0.04  0.03 -0.02 -0.01  0.00  0.00   0   0
## C4  0.10 -0.06 -0.03  0.03 -0.08 -0.02  0.06 -0.05 -0.02 -0.02 -0.01   0   0
## C5  0.02  0.07  0.10  0.01  0.03  0.00 -0.01 -0.01  0.01  0.02  0.01   0   0
## E1  0.03 -0.05 -0.15  0.02  0.03  0.02  0.00 -0.02  0.00  0.02  0.01   0   0
## E2  0.01  0.10  0.06 -0.09  0.01  0.07  0.02  0.05 -0.01  0.00 -0.01   0   0
## E3 -0.14  0.06  0.04 -0.02 -0.02  0.05  0.05  0.00 -0.04  0.03  0.01   0   0
## E4  0.04 -0.06 -0.06  0.03  0.09  0.06  0.00  0.02 -0.03  0.00 -0.01   0   0
## E5  0.01  0.04 -0.07 -0.15  0.01 -0.03  0.04  0.02  0.03 -0.03  0.00   0   0
## N1  0.06 -0.10  0.03 -0.03 -0.01 -0.08 -0.03  0.02 -0.01  0.05 -0.01   0   0
## N2 -0.04  0.02 -0.03  0.07  0.06  0.05  0.03 -0.03 -0.01 -0.03  0.01   0   0
## N3  0.02  0.03  0.02 -0.01 -0.07  0.08  0.02 -0.03  0.05  0.01 -0.01   0   0
## N4  0.00  0.02 -0.01  0.01  0.08 -0.06  0.01  0.05 -0.04 -0.02  0.00   0   0
## N5  0.01  0.05 -0.08 -0.05  0.00 -0.02 -0.06 -0.03  0.00 -0.01  0.01   0   0
## O1 -0.02 -0.01  0.06 -0.04  0.06  0.03 -0.10 -0.05 -0.02  0.00  0.00   0   0
## O2 -0.21 -0.01 -0.05  0.06 -0.06 -0.03 -0.03  0.02  0.00  0.01  0.00   0   0
## O3  0.07 -0.07 -0.04  0.09 -0.01  0.02  0.01  0.06  0.04  0.00  0.01   0   0
## O4 -0.11 -0.13  0.04  0.00  0.00 -0.04  0.02  0.01  0.04  0.00  0.00   0   0
## O5  0.07 -0.12  0.09 -0.08  0.06  0.07  0.00  0.02  0.03  0.00  0.01   0   0
##      h2    u2 com
## A1 0.41 0.59 3.6
## A2 0.65 0.35 5.2
## A3 0.60 0.40 3.7

```

```
## A4 0.48 0.52 5.1
## A5 0.58 0.42 2.8
## C1 0.48 0.52 3.7
## C2 0.61 0.39 4.5
## C3 0.46 0.54 5.0
## C4 0.63 0.37 3.9
## C5 0.62 0.38 4.4
## E1 0.57 0.43 6.0
## E2 0.66 0.34 2.5
## E3 0.58 0.42 3.2
## E4 0.69 0.31 2.9
## E5 0.56 0.44 3.4
## N1 0.77 0.23 2.7
## N2 0.77 0.23 2.8
## N3 0.65 0.35 2.3
## N4 0.62 0.38 3.2
## N5 0.54 0.46 4.6
## O1 0.41 0.59 4.9
## O2 0.42 0.58 4.5
## O3 0.57 0.43 4.0
## O4 0.35 0.65 6.4
## O5 0.45 0.55 4.1
##
##
## MR1 MR2 MR3 MR4 MR5 MR6 MR7 MR8 MR9 MR10 MR11
## SS loadings 4.67 2.41 1.70 1.40 1.07 0.63 0.39 0.34 0.24 0.24 0.21
## Proportion Var 0.19 0.10 0.07 0.06 0.04 0.03 0.02 0.01 0.01 0.01 0.01
## Cumulative Var 0.19 0.28 0.35 0.41 0.45 0.47 0.49 0.50 0.51 0.52 0.53
## Proportion Explained 0.33 0.17 0.12 0.10 0.08 0.04 0.03 0.02 0.02 0.02 0.01
## Cumulative Proportion 0.33 0.50 0.62 0.72 0.80 0.84 0.87 0.89 0.91 0.93 0.94
##
## MR12 MR13 MR14 MR15 MR16 MR17 MR18 MR19 MR20 MR21 MR22
## SS loadings 0.16 0.13 0.12 0.10 0.10 0.07 0.05 0.03 0.02 0.02 0.01
## Proportion Var 0.01 0.01 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00 0.00
## Cumulative Var 0.54 0.54 0.55 0.55 0.56 0.56 0.56 0.56 0.56 0.56 0.56
```

```

## Proportion Explained  0.01 0.01 0.01 0.01 0.01 0.01 0.00 0.00 0.00 0.00 0.00
## Cumulative Proportion 0.95 0.96 0.97 0.98 0.99 0.99 0.99 1.00 1.00 1.00 1.00
##
## MR23 MR24 MR25
## SS loadings          0.00 0.00 0.00
## Proportion Var       0.00 0.00 0.00
## Cumulative Var       0.56 0.56 0.56
## Proportion Explained 0.00 0.00 0.00
## Cumulative Proportion 1.00 1.00 1.00
##
## Mean item complexity = 4
## Test of the hypothesis that 25 factors are sufficient.
##
## df null model = 300 with the objective function = 7.41 with Chi Square = 16484.7
## df of the model are -25 and the objective function was 0
##
## The root mean square of the residuals (RMSR) is 0
## The df corrected root mean square of the residuals is NA
##
## The harmonic n.obs is 2236 with the empirical chi square 0 with prob < NA
## The total n.obs was 2236 with Likelihood Chi Square = 0 with prob < NA
##
## Tucker Lewis Index of factoring reliability = 1.019
## Fit based upon off diagonal values = 1
## Measures of factor score adequacy
##
## MR1 MR2 MR3 MR4 MR5 MR6
## Correlation of (regression) scores with factors 0.96 0.94 0.89 0.86 0.85 0.78
## Multiple R square of scores with factors 0.92 0.88 0.80 0.75 0.72 0.60
## Minimum correlation of possible factor scores 0.85 0.76 0.59 0.49 0.44 0.20
##
## MR7 MR8 MR9 MR10 MR11
## Correlation of (regression) scores with factors 0.70 0.66 0.61 0.59 0.59
## Multiple R square of scores with factors 0.49 0.43 0.37 0.35 0.35
## Minimum correlation of possible factor scores -0.02 -0.13 -0.25 -0.30 -0.29
##
## MR12 MR13 MR14 MR15 MR16

```

```
## Correlation of (regression) scores with factors    0.50  0.47  0.46  0.43  0.42
## Multiple R square of scores with factors          0.25  0.22  0.21  0.19  0.18
## Minimum correlation of possible factor scores     -0.49 -0.57 -0.58 -0.62 -0.64
##                                                    MR17  MR18  MR19  MR20  MR21
## Correlation of (regression) scores with factors    0.39  0.36  0.26  0.23  0.19
## Multiple R square of scores with factors          0.15  0.13  0.07  0.05  0.04
## Minimum correlation of possible factor scores     -0.70 -0.75 -0.86 -0.90 -0.93
##                                                    MR22  MR23  MR24  MR25
## Correlation of (regression) scores with factors    0.17  0.06  0.01    0
## Multiple R square of scores with factors          0.03  0.00  0.00    0
## Minimum correlation of possible factor scores     -0.94 -0.99 -1.00   -1
```

For illustration purpose:

- If using 25 factors (here only 25 items), we can see how loading and proportion of variance decrease with factors;
- The cumulative variance explained will approach 1.

1.2 Screened Model

According to spree plot, take 6 factors.

Note that restrictions on factor number will have practical effect on the model, instead of just trimming the model to the reduced one.

```
fa6_none = fa(currentData, 6, fm='minres', rotate='none')
print(fa6_none)
```

```
## Factor Analysis using method = minres
## Call: fa(r = currentData, nfactors = 6, rotate = "none", fm = "minres")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      MR1  MR2  MR3  MR4  MR5  MR6  h2  u2 com
## A1 -0.22 -0.03  0.15  0.04 -0.42  0.29  0.33  0.67  2.7
## A2  0.46  0.30 -0.20  0.12  0.34 -0.18  0.50  0.50  3.7
## A3  0.52  0.31 -0.25  0.09  0.27  0.02  0.51  0.49  2.8
```

```

## A4  0.40  0.12 -0.15  0.26  0.16  0.03 0.29 0.71 2.7
## A5  0.57  0.18 -0.26  0.03  0.17  0.13 0.47 0.53 2.0
## C1  0.33  0.12  0.44  0.15  0.00  0.10 0.35 0.65 2.4
## C2  0.33  0.18  0.45  0.32  0.06  0.17 0.48 0.52 3.5
## C3  0.32  0.05  0.32  0.33  0.04  0.04 0.32 0.68 3.1
## C4 -0.47  0.13 -0.45 -0.26  0.01  0.26 0.58 0.42 3.4
## C5 -0.49  0.15 -0.26 -0.30  0.10  0.05 0.43 0.57 2.7
## E1 -0.42 -0.20  0.24  0.13  0.25  0.22 0.40 0.60 3.8
## E2 -0.62 -0.06  0.21  0.08  0.32  0.12 0.56 0.44 1.9
## E3  0.55  0.34 -0.09 -0.19 -0.11  0.19 0.51 0.49 2.4
## E4  0.61  0.19 -0.33  0.08 -0.18  0.12 0.56 0.44 2.1
## E5  0.52  0.29  0.11 -0.02 -0.21 -0.03 0.41 0.59 2.1
## N1 -0.43  0.65  0.05  0.10 -0.25 -0.13 0.69 0.31 2.3
## N2 -0.41  0.63  0.11  0.06 -0.20 -0.19 0.67 0.33 2.3
## N3 -0.40  0.61  0.05  0.07  0.00 -0.01 0.54 0.46 1.8
## N4 -0.53  0.41  0.09 -0.04  0.22  0.08 0.51 0.49 2.4
## N5 -0.35  0.41 -0.05  0.20  0.12  0.05 0.35 0.65 2.7
## O1  0.33  0.18  0.24 -0.35  0.00  0.15 0.34 0.66 3.8
## O2 -0.20  0.09 -0.33  0.36 -0.04  0.14 0.31 0.69 3.1
## O3  0.40  0.28  0.19 -0.45  0.01  0.12 0.49 0.51 3.3
## O4 -0.09  0.24  0.18 -0.23  0.29  0.07 0.24 0.76 4.0
## O5 -0.20 -0.05 -0.30  0.42 -0.13  0.20 0.37 0.63 3.2
##
##
##                                     MR1  MR2  MR3  MR4  MR5  MR6
## SS loadings                      4.55 2.30 1.58 1.29 0.96 0.53
## Proportion Var                    0.18 0.09 0.06 0.05 0.04 0.02
## Cumulative Var                    0.18 0.27 0.34 0.39 0.43 0.45
## Proportion Explained              0.41 0.21 0.14 0.12 0.09 0.05
## Cumulative Proportion             0.41 0.61 0.75 0.87 0.95 1.00
##
## Mean item complexity = 2.8
## Test of the hypothesis that 6 factors are sufficient.
##

```

```
## df null model = 300 with the objective function = 7.41 with Chi Square = 16484.7
## df of the model are 165 and the objective function was 0.37
##
## The root mean square of the residuals (RMSR) is 0.02
## The df corrected root mean square of the residuals is 0.03
##
## The harmonic n.obs is 2236 with the empirical chi square 500.78 with prob < 1.6e
## The total n.obs was 2236 with Likelihood Chi Square = 824.5 with prob < 3e-88
##
## Tucker Lewis Index of factoring reliability = 0.926
## RMSEA index = 0.042 and the 90 % confidence intervals are 0.039 0.045
## BIC = -448.05
## Fit based upon off diagonal values = 0.99
## Measures of factor score adequacy
##
## Correlation of (regression) scores with factors MR1 MR2 MR3 MR4 MR5 MR6
## Multiple R square of scores with factors 0.95 0.92 0.86 0.83 0.80 0.71
## Minimum correlation of possible factor scores 0.90 0.84 0.74 0.69 0.65 0.50
## Minimum correlation of possible factor scores 0.80 0.68 0.49 0.38 0.29 0.01
```

However, the old big five model only has 5 factors, and we're to designate this number.

```
fa5_none = fa(currentData, 5, fm='minres', rotate='none')
print(fa5_none)
```

```
## Factor Analysis using method = minres
## Call: fa(r = currentData, nfactors = 5, rotate = "none", fm = "minres")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      MR1  MR2  MR3  MR4  MR5  h2  u2 com
## A1 -0.21 -0.02  0.13  0.04 -0.36 0.20 0.80 2.0
## A2  0.45  0.29 -0.19  0.11  0.32 0.44 0.56 3.2
## A3  0.52  0.31 -0.26  0.09  0.29 0.53 0.47 2.9
## A4  0.40  0.12 -0.16  0.27  0.17 0.30 0.70 2.8
## A5  0.57  0.18 -0.26  0.02  0.18 0.46 0.54 1.9
## C1  0.33  0.12  0.44  0.15  0.00 0.34 0.66 2.3
```

```

## C2  0.32  0.18  0.44  0.32  0.06  0.44  0.56  3.2
## C3  0.32  0.05  0.32  0.34  0.04  0.33  0.67  3.1
## C4 -0.46  0.12 -0.42 -0.25  0.02  0.47  0.53  2.7
## C5 -0.49  0.15 -0.26 -0.31  0.10  0.43  0.57  2.7
## E1 -0.41 -0.20  0.24  0.13  0.24  0.34  0.66  3.1
## E2 -0.62 -0.06  0.21  0.08  0.32  0.55  0.45  1.8
## E3  0.54  0.33 -0.09 -0.19 -0.10  0.46  0.54  2.1
## E4  0.61  0.18 -0.33  0.07 -0.17  0.54  0.46  2.0
## E5  0.52  0.29  0.11 -0.02 -0.22  0.42  0.58  2.1
## N1 -0.43  0.64  0.05  0.10 -0.26  0.68  0.32  2.2
## N2 -0.41  0.62  0.10  0.06 -0.20  0.61  0.39  2.1
## N3 -0.40  0.62  0.05  0.07 -0.01  0.55  0.45  1.8
## N4 -0.53  0.41  0.08 -0.04  0.22  0.51  0.49  2.3
## N5 -0.35  0.42 -0.06  0.20  0.12  0.35  0.65  2.7
## O1  0.33  0.18  0.25 -0.34  0.01  0.32  0.68  3.4
## O2 -0.20  0.09 -0.34  0.35 -0.03  0.28  0.72  2.7
## O3  0.40  0.28  0.20 -0.45  0.02  0.48  0.52  3.1
## O4 -0.09  0.24  0.19 -0.23  0.30  0.24  0.76  3.9
## O5 -0.20 -0.05 -0.30  0.40 -0.12  0.31  0.69  2.7
##
##                                     MR1  MR2  MR3  MR4  MR5
## SS loadings                      4.53 2.28 1.55 1.26 0.92
## Proportion Var                    0.18 0.09 0.06 0.05 0.04
## Cumulative Var                     0.18 0.27 0.33 0.38 0.42
## Proportion Explained              0.43 0.22 0.15 0.12 0.09
## Cumulative Proportion             0.43 0.65 0.79 0.91 1.00
##
## Mean item complexity = 2.6
## Test of the hypothesis that 5 factors are sufficient.
##
## df null model = 300 with the objective function = 7.41 with Chi Square = 16484.7
## df of the model are 185 and the objective function was 0.63
##

```



```
## The root mean square of the residuals (RMSR) is 0.03
## The df corrected root mean square of the residuals is 0.04
##
## The harmonic n.obs is 2236 with the empirical chi square 1046.45 with prob < 3.1
## The total n.obs was 2236 with Likelihood Chi Square = 1400.38 with prob < 1.6e-
##
## Tucker Lewis Index of factoring reliability = 0.878
## RMSEA index = 0.054 and the 90 % confidence intervals are 0.052 0.057
## BIC = -26.42
## Fit based upon off diagonal values = 0.98
## Measures of factor score adequacy
##
## Correlation of (regression) scores with factors
## Multiple R square of scores with factors
## Minimum correlation of possible factor scores
```

	MR1	MR2	MR3	MR4	MR5
Correlation of (regression) scores with factors	0.95	0.91	0.85	0.82	0.79
Multiple R square of scores with factors	0.90	0.83	0.73	0.67	0.63
Minimum correlation of possible factor scores	0.79	0.66	0.45	0.35	0.26

Obviously, with more factors taken into account, more cumulative variance will be explained.

2 Factor Rotation

Factor loadings listed for each variable. The square of it is the variation accounted for.

The hope is that all these correlation coefficients are close to (+/-)1 or 0, so we can easily interpret the meanings of these components (factors). We will accomplish this goal through rotation.

The relationship (correlation coefficient r) between the variable and the factor is shown. We can view it on a **Factor Plot**.

Use 2 factors for illustration purpose, to show how rotation works.

```
fa_unrotated = fa(currentData, nfactores = 2, fm='minres', rotate='none')
# print(fa_unrotated)
```

```
fa_unrotated$loadings #loadings, unrotated
```

```
##
## Loadings:
##      MR1      MR2
## A1 -0.207
## A2  0.439  0.282
## A3  0.511  0.301
## A4  0.394  0.125
## A5  0.563  0.189
## C1  0.311  0.119
## C2  0.300  0.166
## C3  0.308
## C4 -0.439
## C5 -0.473  0.128
## E1 -0.407 -0.200
## E2 -0.610
## E3  0.542  0.340
## E4  0.598  0.190
## E5  0.519  0.296
## N1 -0.432  0.621
## N2 -0.417  0.607
## N3 -0.414  0.626
## N4 -0.539  0.393
## N5 -0.354  0.407
## O1  0.315  0.168
## O2 -0.186
## O3  0.377  0.256
## O4          0.220
## O5 -0.189
##
##              MR1      MR2
## SS loadings    4.392  2.193
```

```
## Proportion Var 0.176 0.088
## Cumulative Var 0.176 0.263
```

Now, rotate it (orthogonal and oblique).

Un-rotated factors are typically not very interpretable (most factors are correlated with many variables), though it has largest variance explained.

Factors are rotated to make them more meaningful and easier to interpret (each variable is associated with a minimal number of factors). Looking for “simple structure”.

The most popular rotational method is Varimax rotations (方差最大旋转). Varimax use orthogonal rotations yielding uncorrelated factors/components.

Varimax attempts to maximize the variance that a single factor can account for (across all variables). This enhances the interpretability of the factors. (kind of simplification of interpretation)

```
fa_otho = fa(currentData,2,fm='minres',rotate='varimax')
print(fa_otho)
```

```
## Factor Analysis using method = minres
## Call: fa(r = currentData, nfactors = 2, rotate = "varimax", fm = "minres")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      MR1   MR2   h2   u2 com
## A1 -0.19  0.09 0.044 0.96 1.4
## A2  0.52  0.01 0.272 0.73 1.0
## A3  0.59 -0.02 0.352 0.65 1.0
## A4  0.40 -0.10 0.171 0.83 1.1
## A5  0.58 -0.14 0.352 0.65 1.1
## C1  0.33 -0.06 0.111 0.89 1.1
## C2  0.34 -0.02 0.118 0.88 1.0
## C3  0.29 -0.12 0.098 0.90 1.3
## C4 -0.32  0.32 0.202 0.80 2.0
## C5 -0.33  0.36 0.240 0.76 2.0
## E1 -0.45  0.05 0.206 0.79 1.0
```

```

## E2 -0.55  0.26 0.377 0.62 1.4
## E3  0.64  0.00 0.409 0.59 1.0
## E4  0.61 -0.16 0.394 0.61 1.1
## E5  0.60 -0.02 0.357 0.64 1.0
## N1 -0.04  0.76 0.572 0.43 1.0
## N2 -0.03  0.74 0.542 0.46 1.0
## N3 -0.02  0.75 0.563 0.44 1.0
## N4 -0.25  0.62 0.444 0.56 1.3
## N5 -0.08  0.53 0.291 0.71 1.0
## O1  0.36 -0.02 0.127 0.87 1.0
## O2 -0.12  0.17 0.041 0.96 1.8
## O3  0.45  0.02 0.207 0.79 1.0
## O4  0.04  0.23 0.056 0.94 1.1
## O5 -0.18  0.06 0.038 0.96 1.2
##
##
##                               MR1  MR2
## SS loadings                   3.77 2.81
## Proportion Var                 0.15 0.11
## Cumulative Var                 0.15 0.26
## Proportion Explained          0.57 0.43
## Cumulative Proportion         0.57 1.00
##
## Mean item complexity = 1.2
## Test of the hypothesis that 2 factors are sufficient.
##
## df null model = 300 with the objective function = 7.41 with Chi Square = 16484.7
## df of the model are 251 and the objective function was 2.73
##
## The root mean square of the residuals (RMSR) is 0.09
## The df corrected root mean square of the residuals is 0.1
##
## The harmonic n.obs is 2236 with the empirical chi square 10195.98 with prob < 0
## The total n.obs was 2236 with Likelihood Chi Square = 6068.95 with prob < 0

```

```
##
## Tucker Lewis Index of factoring reliability = 0.57
## RMSEA index = 0.102 and the 90 % confidence intervals are 0.1 0.104
## BIC = 4133.13
## Fit based upon off diagonal values = 0.83
## Measures of factor score adequacy
##
## Correlation of (regression) scores with factors    MR1  MR2
## Multiple R square of scores with factors          0.84 0.84
## Minimum correlation of possible factor scores      0.67 0.67
```

Notice that the loading matrix changes, with values moving aside either towards 0 or 1. However, the communality doesn't change much, and this result is quite natural.

As for rotation, Oblimin is the default in R, which is an oblique rotation. Varimax is the most popular orthogonal rotation. Other rotational methods include Quartimax (Orthogonal, each variable accounted by minimal number of factors, 简化对变量的解释), Equamax (Orthogonal), Promax (oblique, this one is fast!). Oblique rotation yields correlated factors.

```
fa_otho$loadings # show the loadings after rotation
```

```
##
## Loadings:
##      MR1      MR2
## A1 -0.189
## A2 0.521
## A3 0.593
## A4 0.401 -0.104
## A5 0.577 -0.139
## C1 0.327
## C2 0.342
## C3 0.289 -0.120
## C4 -0.319 0.317
```

```

## C5 -0.332  0.359
## E1 -0.451
## E2 -0.554  0.265
## E3  0.639
## E4  0.608 -0.157
## E5  0.597
## N1          0.756
## N2          0.736
## N3          0.750
## N4 -0.248  0.619
## N5          0.533
## O1  0.356
## O2 -0.115  0.167
## O3  0.455
## O4          0.233
## O5 -0.183
##
##              MR1   MR2
## SS loadings    3.771 2.814
## Proportion Var 0.151 0.113
## Cumulative Var 0.151 0.263

fa_otho$rot.mat  # show the rotation matrix

##           [,1]      [,2]
## [1,] 0.8470713 -0.5314793
## [2,] 0.5314793  0.8470713

fa_oblique = fa(currentData,2,fm='minres',rotate='promax')
print(fa_oblique)

## Factor Analysis using method = minres
## Call: fa(r = currentData, nfactors = 2, rotate = "promax", fm = "minres")
## Standardized loadings (pattern matrix) based upon correlation matrix
##      MR1   MR2   h2   u2 com

```

```

## A1 -0.18  0.06 0.044 0.96 1.2
## A2  0.55  0.11 0.272 0.73 1.1
## A3  0.62  0.10 0.352 0.65 1.1
## A4  0.40 -0.03 0.171 0.83 1.0
## A5  0.58 -0.03 0.352 0.65 1.0
## C1  0.33  0.00 0.111 0.89 1.0
## C2  0.36  0.05 0.118 0.88 1.0
## C3  0.28 -0.07 0.098 0.90 1.1
## C4 -0.27  0.27 0.202 0.80 2.0
## C5 -0.28  0.31 0.240 0.76 2.0
## E1 -0.47 -0.04 0.206 0.79 1.0
## E2 -0.53  0.17 0.377 0.62 1.2
## E3  0.68  0.13 0.409 0.59 1.1
## E4  0.61 -0.04 0.394 0.61 1.0
## E5  0.63  0.10 0.357 0.64 1.0
## N1  0.12  0.79 0.572 0.43 1.0
## N2  0.12  0.77 0.542 0.46 1.0
## N3  0.14  0.79 0.563 0.44 1.1
## N4 -0.13  0.60 0.444 0.56 1.1
## N5  0.02  0.55 0.291 0.71 1.0
## O1  0.37  0.05 0.127 0.87 1.0
## O2 -0.09  0.15 0.041 0.96 1.6
## O3  0.48  0.11 0.207 0.79 1.1
## O4  0.09  0.25 0.056 0.94 1.3
## O5 -0.18  0.03 0.038 0.96 1.1
##
##
##                                     MR1  MR2
## SS loadings                      3.81 2.78
## Proportion Var                   0.15 0.11
## Cumulative Var                    0.15 0.26
## Proportion Explained              0.58 0.42
## Cumulative Proportion             0.58 1.00
##

```

```

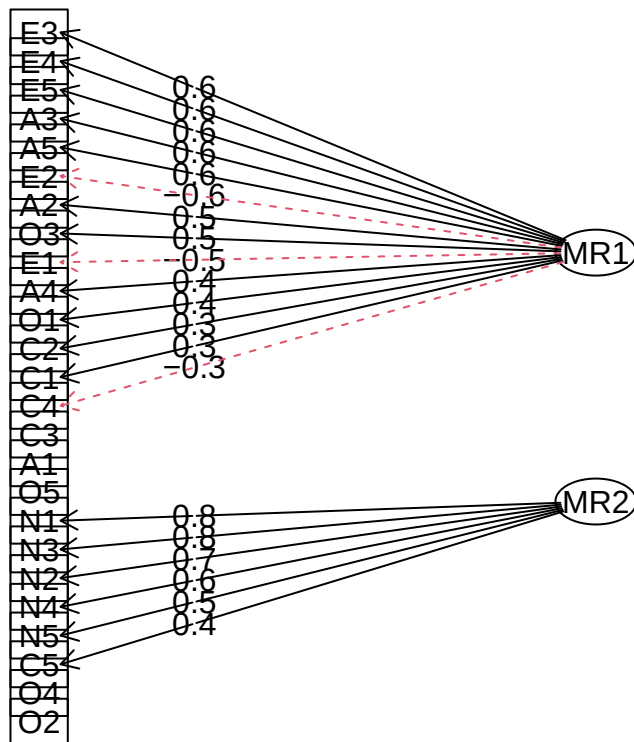
## With factor correlations of
##      MR1   MR2
## MR1  1.00 -0.37
## MR2 -0.37  1.00
##
## Mean item complexity = 1.2
## Test of the hypothesis that 2 factors are sufficient.
##
## df null model = 300 with the objective function = 7.41 with Chi Square = 16484.7
## df of the model are 251 and the objective function was 2.73
##
## The root mean square of the residuals (RMSR) is 0.09
## The df corrected root mean square of the residuals is 0.1
##
## The harmonic n.obs is 2236 with the empirical chi square 10195.98 with prob < 0
## The total n.obs was 2236 with Likelihood Chi Square = 6068.95 with prob < 0
##
## Tucker Lewis Index of factoring reliability = 0.57
## RMSEA index = 0.102 and the 90 % confidence intervals are 0.1 0.104
## BIC = 4133.13
## Fit based upon off diagonal values = 0.83
## Measures of factor score adequacy
##
##                                     MR1  MR2
## Correlation of (regression) scores with factors 0.92 0.92
## Multiple R square of scores with factors         0.85 0.85
## Minimum correlation of possible factor scores    0.70 0.70

```

Use diagram to show how variables relate to the factors.

```
fa.diagram(fa_otho)
```


Factor Analysis

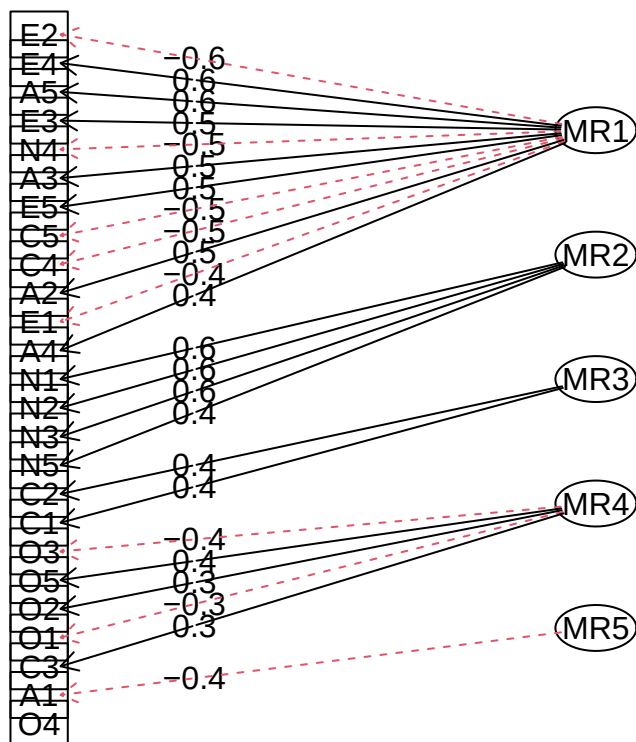


What if we use 5 factors, either rotated or not. To see whether big five really works here.

Note `fa.diagram` use `r = 0.3` as a cut off, you can change it.

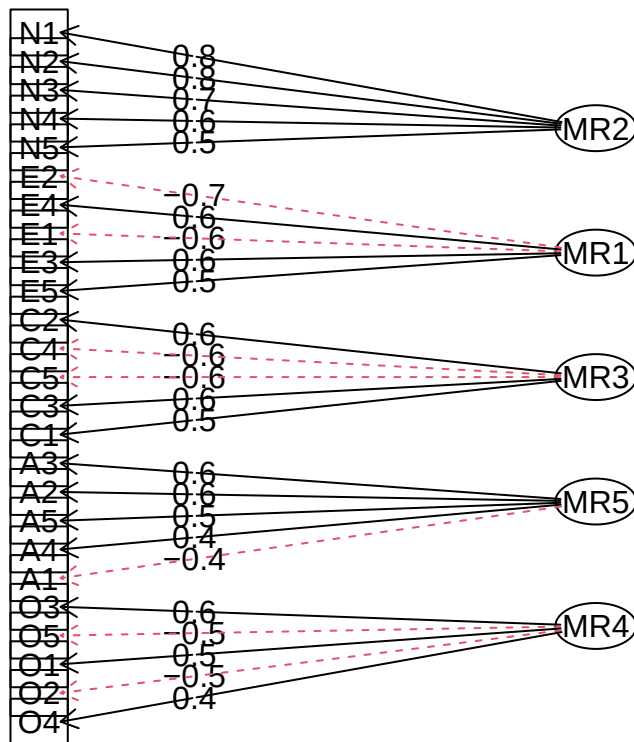
```
fa5_none = fa(currentData,5,fm='minres',rotate='none')
fa.diagram(fa5_none)
```

Factor Analysis



```
fa5_otho = fa(currentData,5,fm='minres',rotate='varimax')
fa.diagram(fa5_otho)
```

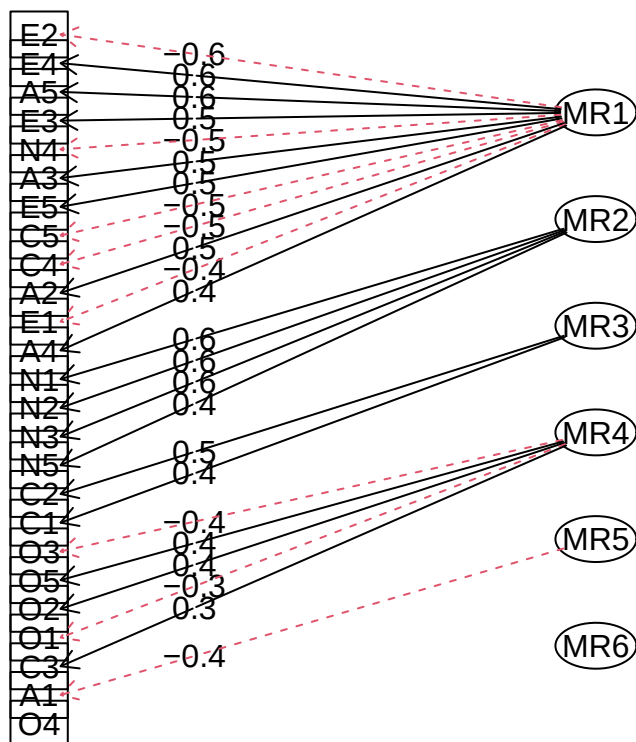
Factor Analysis



Interestingly, the 6-factor model does not work that well.

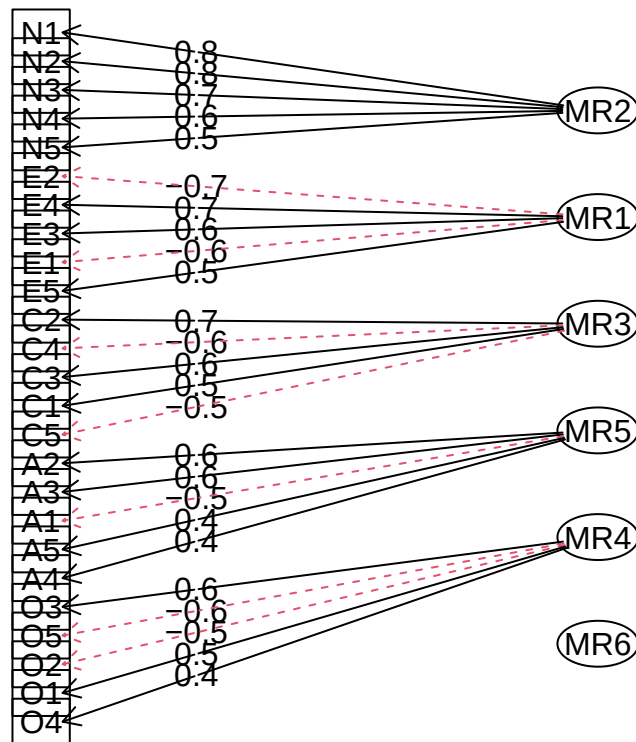
```
fa6_none = fa(currentData, 6, fm='minres', rotate='none')
fa.diagram(fa6_none)
```

Factor Analysis



```
fa6_otho = fa(currentData,6,fm='minres',rotate='varimax')
fa.diagram(fa6_otho)
```

Factor Analysis



Scores, or the coefficient for the model: $\text{Factors} = \text{scores} * X$ where X are the original data, Factors are the obtained latent variables, scores are their coefficients.

The scores here are for normalized data, which is the case for SPSS.

```
head(fa_otho$scores)
```

```
##           MR1           MR2
## 1  1.27436156 -0.05162498
## 2 -2.25649076  0.45934922
## 3 -0.09521188 -0.12571634
## 4 -0.70689909 -0.56922445
## 5 -0.18784942 -0.75916140
## 6  0.91231347  1.16628979
```

The below is for you to hand check the meaning of SS loading, uniqueness and communality.

Communality, the variance that can be explained by factors, which should be high if the model is good.

```
# communality
```

```
apply(fa_otho$loadings^2,1,sum)
```

```
##          A1          A2          A3          A4          A5          C1          C2
## 0.04359507 0.27197456 0.35182910 0.17116362 0.35236109 0.11106610 0.11760829
##          C3          C4          C5          E1          E2          E3          E4
## 0.09774264 0.20204612 0.23964636 0.20559167 0.37742115 0.40892440 0.39419870
##          E5          N1          N2          N3          N4          N5          O1
## 0.35693755 0.57226220 0.54214564 0.56338639 0.44421810 0.29146687 0.12733686
##          O2          O3          O4          O5
## 0.04122094 0.20719293 0.05598365 0.03767935
```

```
# uniqueness
```

```
1 - apply(fa_otho$loadings^2,1,sum)
```

```
##          A1          A2          A3          A4          A5          C1          C2          C3
## 0.9564049 0.7280254 0.6481709 0.8288364 0.6476389 0.8889339 0.8823917 0.9022574
##          C4          C5          E1          E2          E3          E4          E5          N1
## 0.7979539 0.7603536 0.7944083 0.6225788 0.5910756 0.6058013 0.6430625 0.4277378
##          N2          N3          N4          N5          O1          O2          O3          O4
## 0.4578544 0.4366136 0.5557819 0.7085331 0.8726631 0.9587791 0.7928071 0.9440163
##          O5
## 0.9623207
```

```
P = EFA(data = currentData,vars = names(currentData))
```

```
##
```

```
## Principal Component Analysis
```

```
##
```

```
## Summary:
```

```
## Total Items: 25
```

```

## Scale Range: 1 ~ 6
## Total Cases: 2236
## Valid Cases: 2236 (100.0%)
##
## Extraction Method:
## - Principal Component Analysis
## Rotation Method:
## - Varimax (with Kaiser Normalization)
##
## KMO and Bartlett's Test:
## - Kaiser-Meyer-Olkin (KMO) Measure of Sampling Adequacy: MSA = 0.847
## - Bartlett's Test of Sphericity: Approx.  $\chi^2(300) = 16484.78$ ,  $p < 1e-99$  ***
##
## Total Variance Explained:
##
##              Eigenvalue Variance % Cumulative % SS Loading Variance % Cumulative %
##
## Component 1      5.069      20.274      20.274      3.089      12.356      12.356
## Component 2      2.762      11.050      31.324      2.596      10.385      22.741
## Component 3      2.153       8.610      39.934      2.570      10.280      33.021
## Component 4      1.892       7.569      47.504      2.497       9.989      43.010
## Component 5      1.518       6.070      53.574      2.092       8.368      51.379
## Component 6      1.079       4.315      57.889      1.628       6.511      57.889
## Component 7       0.831       3.324      61.213
## Component 8       0.805       3.218      64.431
## Component 9       0.714       2.856      67.287
## Component 10      0.702       2.806      70.093
## Component 11      0.681       2.723      72.817
## Component 12      0.649       2.596      75.413
## Component 13      0.631       2.525      77.938
## Component 14      0.588       2.352      80.290
## Component 15      0.566       2.264      82.554
## Component 16      0.545       2.179      84.733

```

```

## Component 17      0.520      2.080      86.813
## Component 18      0.494      1.975      88.788
## Component 19      0.483      1.931      90.719
## Component 20      0.443      1.770      92.489
## Component 21      0.429      1.715      94.205
## Component 22      0.407      1.628      95.833
## Component 23      0.389      1.556      97.389
## Component 24      0.385      1.539      98.928
## Component 25      0.268      1.072     100.000
##
##
## Component Loadings (Rotated) (Sorted by Size):
##
##          RC2      RC5      RC3      RC6      RC1      RC4  Communality
##
## N1   0.836 -0.158 -0.041 -0.098 -0.026  0.065      0.741
## N2   0.835 -0.149 -0.021 -0.078 -0.033 -0.038      0.728
## N3   0.794 -0.005 -0.049  0.088  0.043  0.046      0.644
## N4   0.609 -0.027 -0.169  0.432  0.064 -0.002      0.591
## N5   0.609  0.147 -0.033  0.211 -0.096  0.188      0.483
## A2   0.058  0.740  0.117 -0.156  0.026 -0.112      0.602
## A3  -0.023  0.708  0.089 -0.184  0.224  0.079      0.600
## A1   0.072 -0.636  0.092 -0.064  0.229  0.436      0.664
## A5  -0.185  0.594  0.061 -0.227  0.322  0.165      0.574
## A4  -0.088  0.556  0.239 -0.111 -0.003  0.201      0.427
## C2   0.075  0.103  0.730  0.072  0.192  0.059      0.595
## C4   0.193 -0.072 -0.686  0.182  0.152  0.314      0.668
## C3  -0.032  0.125  0.678  0.005  0.003  0.045      0.478
## C1   0.014  0.006  0.650  0.014  0.253 -0.069      0.491
## C5   0.280 -0.067 -0.625  0.240  0.087  0.030      0.539
## E2   0.209 -0.192 -0.071  0.724 -0.157  0.033      0.636
## E1  -0.054 -0.211  0.092  0.720 -0.054  0.166      0.605
## E4  -0.148  0.388  0.080 -0.559  0.258  0.262      0.626

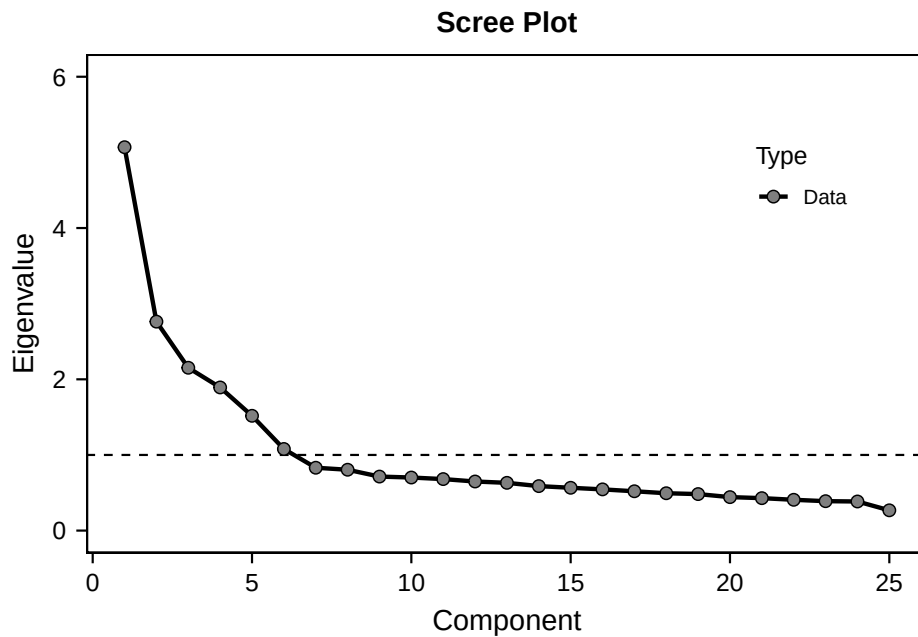
```



```

## E5    0.072  0.127  0.334 -0.518  0.332 -0.070      0.517
## 04    0.201  0.157 -0.047  0.434  0.405 -0.243      0.478
## 01   -0.070 -0.010  0.124 -0.049  0.683 -0.199      0.529
## 03    0.001  0.094  0.057 -0.174  0.669 -0.307      0.585
## E3   -0.016  0.274  0.071 -0.416  0.586  0.069      0.601
## 05    0.042 -0.017 -0.043  0.015 -0.261  0.699      0.561
## 02    0.164  0.134 -0.105  0.040 -0.204  0.641      0.510
##
## Communality = Sum of Squared (SS) Factor Loadings
## (Uniqueness = 1 - Communality)

```



P\$eigenvalues

```

##          Eigenvalue Variance % Cumulative % SS Loading Variance %
## Component 1      5.0685162  20.274065      20.27406   3.089101  12.356404
## Component 2      2.7624793  11.049917      31.32398   2.596271  10.385084
## Component 3      2.1526230   8.610492      39.93447   2.569883  10.279533
## Component 4      1.8923330   7.569332      47.50381   2.497253   9.989012
## Component 5      1.5175329   6.070132      53.57394   2.092123   8.368493

```

## Component 6	1.0788293	4.315317	57.88925	1.627682	6.510729
## Component 7	0.8309057	3.323623	61.21288	NA	NA
## Component 8	0.8045002	3.218001	64.43088	NA	NA
## Component 9	0.7140883	2.856353	67.28723	NA	NA
## Component 10	0.7015381	2.806152	70.09338	NA	NA
## Component 11	0.6808421	2.723368	72.81675	NA	NA
## Component 12	0.6489735	2.595894	75.41265	NA	NA
## Component 13	0.6312563	2.525025	77.93767	NA	NA
## Component 14	0.5880320	2.352128	80.28980	NA	NA
## Component 15	0.5659652	2.263861	82.55366	NA	NA
## Component 16	0.5448396	2.179358	84.73302	NA	NA
## Component 17	0.5199335	2.079734	86.81275	NA	NA
## Component 18	0.4938686	1.975474	88.78823	NA	NA
## Component 19	0.4827362	1.930945	90.71917	NA	NA
## Component 20	0.4425003	1.770001	92.48917	NA	NA
## Component 21	0.4288706	1.715483	94.20466	NA	NA
## Component 22	0.4070974	1.628390	95.83305	NA	NA
## Component 23	0.3888753	1.555501	97.38855	NA	NA
## Component 24	0.3847626	1.539050	98.92760	NA	NA
## Component 25	0.2681008	1.072403	100.00000	NA	NA
##	Cumulative %				
## Component 1	12.35640				
## Component 2	22.74149				
## Component 3	33.02102				
## Component 4	43.01003				
## Component 5	51.37853				
## Component 6	57.88925				
## Component 7	NA				
## Component 8	NA				
## Component 9	NA				
## Component 10	NA				
## Component 11	NA				
## Component 12	NA				

```
## Component 13      NA
## Component 14      NA
## Component 15      NA
## Component 16      NA
## Component 17      NA
## Component 18      NA
## Component 19      NA
## Component 20      NA
## Component 21      NA
## Component 22      NA
## Component 23      NA
## Component 24      NA
## Component 25      NA
```

```
P$loadings
```

```
##          RC2          RC5          RC3          RC6          RC1          RC4
## N1  0.836491887 -0.157590680 -0.04108418 -0.097998494 -0.025939731  0.06465551
## N2  0.834508658 -0.149156942 -0.02064085 -0.077864197 -0.032563963 -0.03816179
## N3  0.793711537 -0.005168488 -0.04888555  0.088043491  0.042752044  0.04602475
## N4  0.609322682 -0.027472354 -0.16871044  0.431528485  0.063980877 -0.00162981
## N5  0.609055668  0.147324483 -0.03340184  0.210953824 -0.095819511  0.18800144
## A2  0.057524885  0.739900333  0.11664100 -0.156447169  0.026011852 -0.11189172
## A3 -0.022773080  0.708278364  0.08884781 -0.183650315  0.224099530  0.07914837
## A1  0.072027071 -0.635726330  0.09156661 -0.063644039  0.229021008  0.43554813
## A5 -0.185158668  0.594493141  0.06074520 -0.227027404  0.322347467  0.16497047
## A4 -0.087699627  0.556153428  0.23937838 -0.110857744 -0.003473359  0.20080148
## C2  0.074779715  0.103231141  0.73004958  0.072391718  0.192269544  0.05935875
## C4  0.192541610 -0.072229949 -0.68596718  0.181892561  0.151801380  0.31422451
## C3 -0.031689921  0.125088185  0.67775704  0.005040871  0.003201663  0.04546638
## C1  0.013508483  0.006478096  0.64950421  0.013871224  0.252676838 -0.06868019
## C5  0.279840593 -0.067415850 -0.62465545  0.240249583  0.086582100  0.03010505
## E2  0.208799741 -0.191702723 -0.07061461  0.724424287 -0.157433230  0.03295199
## E1 -0.054138742 -0.211222539  0.09164581  0.720337059 -0.053606737  0.16560687
```

```
## E4 -0.147608529  0.387783331  0.07957953 -0.559071760  0.258448370  0.26180636
## E5  0.072113100  0.126769699  0.33384919 -0.518452171  0.332493323 -0.06969270
## O4  0.200561610  0.156760944 -0.04737357  0.434084755  0.404647132 -0.24277789
## O1 -0.069524090 -0.009900085  0.12373491 -0.048918290  0.683065226 -0.19916320
## O3  0.001108861  0.093768892  0.05719735 -0.174044696  0.669355360 -0.30726619
## E3 -0.016284826  0.273872428  0.07112138 -0.415697273  0.586162005  0.06888588
## O5  0.042186541 -0.016811945 -0.04314908  0.015058710 -0.261138655  0.69879394
## O2  0.164325447  0.134322459 -0.10524777  0.039920315 -0.203755777  0.64079428
##      Communality
## N1  0.7406983
## N2  0.7276581
## N3  0.6440922
## N4  0.5908051
## N5  0.4827964
## A2  0.6020388
## A3  0.6002833
## A1  0.6639237
## A5  0.5740604
## A4  0.4269226
## C2  0.5949527
## C4  0.6677060
## C3  0.4781088
## C1  0.4908351
## C5  0.5391727
## E2  0.6359953
## E1  0.6051297
## E4  0.6263965
## E5  0.5169277
## O4  0.4781532
## O1  0.5288790
## O3  0.5848060
## E3  0.6014649
## O5  0.5606573
```

```
## 02    0.5098498
```

```
P$result
```

```
## Principal Components Analysis
```

```
## Call: psych::principal(r = data, nfactors = nfactors, rotate = rotation)
```

```
## Standardized loadings (pattern matrix) based upon correlation matrix
```

```
##      RC2   RC5   RC3   RC6   RC1   RC4   h2   u2 com
## A1  0.07 -0.64  0.09 -0.06  0.23  0.43  0.66  0.34  2.2
## A2  0.06  0.74  0.12 -0.16  0.03 -0.11  0.60  0.40  1.2
## A3 -0.02  0.71  0.09 -0.18  0.23  0.08  0.60  0.40  1.4
## A4 -0.09  0.56  0.24 -0.11  0.00  0.20  0.43  0.57  1.8
## A5 -0.19  0.59  0.06 -0.22  0.32  0.16  0.57  0.43  2.3
## C1  0.01  0.01  0.65  0.02  0.25 -0.07  0.49  0.51  1.3
## C2  0.07  0.10  0.73  0.07  0.19  0.06  0.59  0.41  1.2
## C3 -0.03  0.12  0.68  0.01  0.00  0.05  0.48  0.52  1.1
## C4  0.19 -0.07 -0.69  0.18  0.15  0.31  0.67  0.33  1.9
## C5  0.28 -0.07 -0.62  0.24  0.09  0.03  0.54  0.46  1.8
## E1 -0.05 -0.21  0.09  0.72 -0.06  0.16  0.61  0.39  1.3
## E2  0.21 -0.19 -0.07  0.72 -0.16  0.03  0.64  0.36  1.5
## E3 -0.02  0.27  0.07 -0.41  0.59  0.07  0.60  0.40  2.3
## E4 -0.15  0.39  0.08 -0.56  0.26  0.26  0.63  0.37  3.0
## E5  0.07  0.13  0.33 -0.52  0.34 -0.07  0.52  0.48  2.8
## N1  0.84 -0.16 -0.04 -0.10 -0.03  0.06  0.74  0.26  1.1
## N2  0.83 -0.15 -0.02 -0.08 -0.03 -0.04  0.73  0.27  1.1
## N3  0.79 -0.01 -0.05  0.09  0.04  0.05  0.64  0.36  1.0
## N4  0.61 -0.03 -0.17  0.43  0.06  0.00  0.59  0.41  2.0
## N5  0.61  0.15 -0.03  0.21 -0.10  0.19  0.48  0.52  1.7
## O1 -0.07 -0.01  0.12 -0.05  0.68 -0.20  0.53  0.47  1.3
## O2  0.16  0.13 -0.11  0.04 -0.20  0.64  0.51  0.49  1.5
## O3  0.00  0.09  0.06 -0.17  0.67 -0.31  0.58  0.42  1.6
## O4  0.20  0.16 -0.05  0.44  0.40 -0.25  0.48  0.52  3.4
## O5  0.04 -0.02 -0.04  0.01 -0.26  0.70  0.56  0.44  1.3
##
```

```

##          RC2  RC5  RC3  RC6  RC1  RC4
## SS loadings      3.09 2.60 2.57 2.49 2.10 1.63
## Proportion Var   0.12 0.10 0.10 0.10 0.08 0.07
## Cumulative Var   0.12 0.23 0.33 0.43 0.51 0.58
## Proportion Explained 0.21 0.18 0.18 0.17 0.14 0.11
## Cumulative Proportion 0.21 0.39 0.57 0.74 0.89 1.00
##
## Mean item complexity = 1.7
## Test of the hypothesis that 6 components are sufficient.
##
## The root mean square of the residuals (RMSR) is 0.05
## with the empirical chi square 3727.83 with prob < 0
##
## Fit based upon off diagonal values = 0.94
P$result.kaiser

##
## Call: NULL
## Standardized loadings (pattern matrix) based upon correlation matrix
##      RC2  RC5  RC3  RC6  RC1  RC4  h2  u2
## A1  0.07 -0.64  0.09 -0.06  0.23  0.44 0.66 0.34
## A2  0.06  0.74  0.12 -0.16  0.03 -0.11 0.60 0.40
## A3 -0.02  0.71  0.09 -0.18  0.22  0.08 0.60 0.40
## A4 -0.09  0.56  0.24 -0.11  0.00  0.20 0.43 0.57
## A5 -0.19  0.59  0.06 -0.23  0.32  0.16 0.57 0.43
## C1  0.01  0.01  0.65  0.01  0.25 -0.07 0.49 0.51
## C2  0.07  0.10  0.73  0.07  0.19  0.06 0.59 0.41
## C3 -0.03  0.13  0.68  0.01  0.00  0.05 0.48 0.52
## C4  0.19 -0.07 -0.69  0.18  0.15  0.31 0.67 0.33
## C5  0.28 -0.07 -0.62  0.24  0.09  0.03 0.54 0.46
## E1 -0.05 -0.21  0.09  0.72 -0.05  0.17 0.61 0.39
## E2  0.21 -0.19 -0.07  0.72 -0.16  0.03 0.64 0.36
## E3 -0.02  0.27  0.07 -0.42  0.59  0.07 0.60 0.40

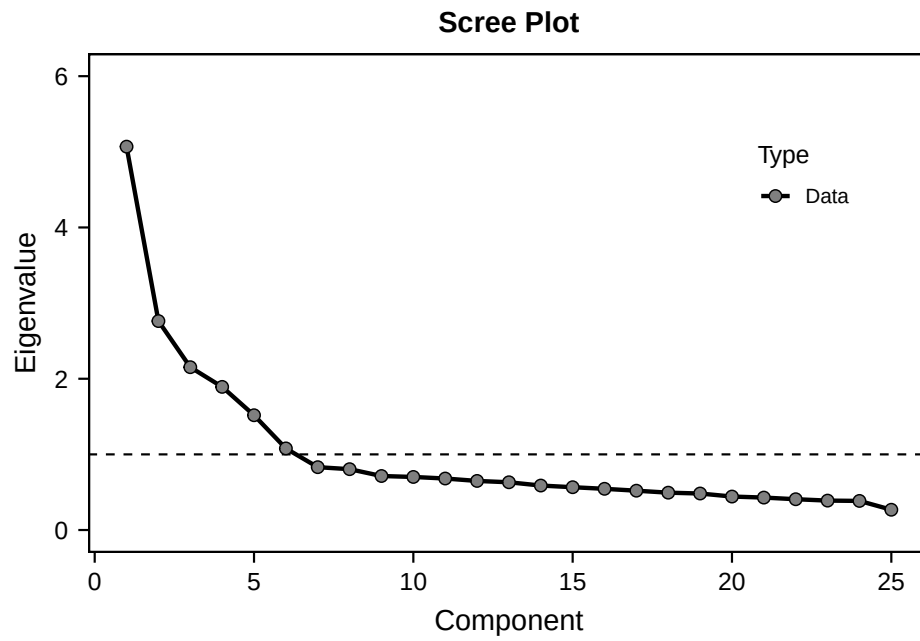
```

```

## E4 -0.15  0.39  0.08 -0.56  0.26  0.26  0.63  0.37
## E5  0.07  0.13  0.33 -0.52  0.33 -0.07  0.52  0.48
## N1  0.84 -0.16 -0.04 -0.10 -0.03  0.06  0.74  0.26
## N2  0.83 -0.15 -0.02 -0.08 -0.03 -0.04  0.73  0.27
## N3  0.79 -0.01 -0.05  0.09  0.04  0.05  0.64  0.36
## N4  0.61 -0.03 -0.17  0.43  0.06  0.00  0.59  0.41
## N5  0.61  0.15 -0.03  0.21 -0.10  0.19  0.48  0.52
## O1 -0.07 -0.01  0.12 -0.05  0.68 -0.20  0.53  0.47
## O2  0.16  0.13 -0.11  0.04 -0.20  0.64  0.51  0.49
## O3  0.00  0.09  0.06 -0.17  0.67 -0.31  0.58  0.42
## O4  0.20  0.16 -0.05  0.43  0.40 -0.24  0.48  0.52
## O5  0.04 -0.02 -0.04  0.02 -0.26  0.70  0.56  0.44
##
##
##              RC2  RC5  RC3  RC6  RC1  RC4
## SS loadings      3.09 2.60 2.57 2.50 2.09 1.63
## Proportion Var    0.12 0.10 0.10 0.10 0.08 0.07
## Cumulative Var     0.12 0.23 0.33 0.43 0.51 0.58
## Proportion Explained 0.21 0.18 0.18 0.17 0.14 0.11
## Cumulative Proportion 0.21 0.39 0.57 0.74 0.89 1.00

```

```
P$scree.plot
```



3 Appendix

3.1 `psych::describe()`